

LISA SHI

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EDUCATION

University of British Columbia

M.Eng. in Mechanical Engineering

Coursework: Linear FEA, Thermodynamics, Biomechanics, Biofluids, Linear Vibrations

Research: Elastic properties of biomembranes

Vancouver, BC

Anticipated: Aug. 2027

B.A.Sc. in Engineering Physics — GPA: 3.7/4.0

Sep. 2021 – May 2026

Awards: Engineering Scholarship, Trek Excellence Scholarship, Dean's Honour List

Coursework: Continuum & Solid Mechanics, Machine Design, CAM, Modern Control Theory

WORK EXPERIENCE

Phase Analyzer Company

R&D Engineering Co-op

Richmond, BC

May 2024 – Dec. 2024

- Supported mechanical development of a precision fluid measurement system through component design, system integration, and performance optimization.
- Researched and evaluated cryogenic insulation materials for operation down to **-120°C**, improving thermal management of petroleum measurement systems.
- Designed and validated **CNC-machined assembly jigs and custom fixtures** using **SOLIDWORKS**, improving assembly precision and repeatability while reducing assembly time.
- Conducted quality inspections and prepared non-conformance documentation to **ISO9001 requirements** for **50+ components**, identifying manufacturing and dimensional issues.
- Collaborated with suppliers, engineers, and assembly teams to coordinate part sourcing, manufacturing processes, and quality requirements.

Primex Manufacturing

Product Designer Co-op

Langley, BC

Jan. 2023 – May 2023

- Designed and improved **10 thermoplastic injection-molded fiber optic enclosures**, optimizing structural performance and manufacturability for telecommunications applications.
- Created **50+ technical drawings** with **GD&T** specifications in **SOLIDWORKS**, supporting accurate manufacturing and production hand-off.
- Managed revision control for **200+ engineering files** using **SOLIDWORKS PDM**, maintaining configuration accuracy across design iterations.
- Integrated fiber optic cable routing and connector interfaces into mechanical housings to improve installation efficiency.

PROJECTS

High-Proficiency Piano-Playing Robot

Sep. 2024 – Apr. 2026

- Designed a robotic finger mechanism to replicate human-like dynamics, articulation, and timing for piano performance.
- Designed and prototyped a custom voice-coil actuator in **OnShape** using **3D-printed and waterjet-cut components**, optimizing modular assembly, mechanical noise, and heat dissipation.
- Designed a lateral-motion actuator and characterized stepper motor performance using open-loop control with an **ESP32** microcontroller.

Nano-Satellite — UBC Orbit & European Space Agency Fly Your Satellite Program

Apr. 2024 – Sep. 2025

- Performed structural **FEA** in **ANSYS Mechanical** to evaluate satellite response to natural-frequency resonance, static loads, and random vibration during launch and orbital operation.
- Designed a **20G-rated camera mount** and developed **30+ mechanical models** in **OnShape** for satellite integration.

TECHNICAL SKILLS

Mechanical/Structural: SOLIDWORKS, OnShape, AutoCAD, Siemens NX, ANSYS

Materials/Fabrication: 3D Printing, Laser Cutting, Waterjet Cutting, Hand Soldering